

Quantum-Inspired Cyber Threat Detection for Digital Signature Security

PROBLEM STATEMENT SIH26141 · EGREEN QUANTA LLP · SMART INDIA HACKATHON 2026

3904 passed 0 failed, 0 skipped	12,494 trials 7 experiments	3.3964e-10 proven false-positive bound	499 doctests inside the suite	0 lines of ML constraint D4
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Nothing here is a claim about coverage. Each row carries the figure that settles it, and every figure regenerates from a recorded seed and a committed command, which is the project's ninth convention. Where the honest answer is narrower than the brief, the row says so rather than rounding up.

THE DELIVERY TABLE, ROW BY ROW

#	ASKED FOR	WHAT SHIPS	THE FIGURE THAT SETTLES IT
1	Mathematical model of teleportation-based QDS	core/ states, teleport, paulis, measure, with docs/MODELLING.md	Bell-state entanglement, teleportation, Pauli correction and projective rules: the brief names four components and all four are implemented, not described
2	Quantum-inspired threat detection framework	sih141/detect/ with nine hypotheses across three threshold families	Every threshold derived, none tuned against attack data. Family-wise correction gives a proven false-positive bound of 3.3964e-10
3	Signature generation and verification	protocol/ distribute, keys, signature, verify	340/340 honest verifier-parties accept, from L=192 up to L=12288. No legitimate signature was ever rejected
4	Attack simulation module	sih141/attacks/, six adversaries	The brief names four scenarios and asks them to be simulated; the sweep ran 12,494 trials across seven experiments in 23.6 minutes
6	Prototype: simulation environment, verification interface, threat detection dashboard, logging of security events	sih141/web/ and sih141/audit.py	All four components. A click returns a verdict in 0.41 s at the default length; the event log and its endpoint landed in commit ad1147f

The brief's numbering runs 1, 2, 3, 4, 6: there is no row 5. Security analysis and performance evaluation, named in its Expected Solution without rows of their own, are docs/SECURITY.md and the Phase 5 sweep.

THE FIVE OBJECTIVES

OBJECTIVE	WHERE IT STANDS
Design a quantum-inspired threat detection framework for teleportation-based QDS	Built, with no machine learning anywhere in the detection path. That is the brief's hard constraint and this project's convention D4: an ML detector gives an empirical rate with no provable bound, discarding the guarantee that motivates QDS at all
Detect forgery, impersonation, replay, and unauthorized verification attempts	Forgery and replay at 40/40 . Impersonation at the same rate for either seam alone, and excluded by assumption (AUTH) when both are held at once. Unauthorized verification decomposes into three routes and all three are answered; the row below the tables says exactly how
Use Pauli eigenstates, measurement analysis, and statistical threshold methods	Hoeffding, multiplicative Chernoff, exact binomial tails, McDiarmid for CHSH, Wilson intervals on every measured rate. Bases are X, Y and Z, so a matched position occurs one time in three
Efficient verification that keeps information-theoretic security	Four lines of arithmetic behind six admissibility checks. 0.41 s per run at L=192, near 2.3 s at the live ceiling of L=1024
Evaluate through forgery probability analysis, attack simulation, and performance metrics	Closed form and simulation agree on 21 of 21 rows, the exact in-model probability landing inside the 99% Wilson interval every time

BUILT WITHOUT BEING ASKED

WHAT	WHY IT IS HERE	EVIDENCE
Count starvation	A fifth attack surface the brief never names, found by this project's own Phase 2 audit. The matched count crossing the wire is an unauthenticated integer chosen by a party the threat model already distrusts	denial 20/20 , deterministic and free, one integer. Detected 40/40
Forgery split into two adversaries	The brief writes one word. An outsider who holds nothing and a recipient who holds half the evidence are not the same attacker, and pooling them publishes a rate describing neither	bounds five orders apart: 3.17e-07 against 1.12e-12 at $L=192$
Repudiation and non-repudiation	Never mentioned in the brief, and one of the three properties that make a signature a signature. Phase A-prime symmetrisation is what buys it	repudiation curve over eleven lengths, with the pooled matched-count floor
ROC and noise calibration	A detector quoted at one operating point tells you nothing about the others. Six noise levels, and a null the tables mark as noiseless wherever it is	roc.svg, noise.md, and a null_is_noiseless column on every table
Reproducibility as a rule, not a habit	Convention D9. Every published number regenerates from a recorded seed and a committed command. The Phase 5 sweep was thrown away and recomputed once because its manifests recorded a dirty tree	seven tables, each printing the command that rebuilds it. All rates identical on recompute; both charts byte-identical
Documentation that fails the build when it lies	499 of the suite's tests are doctests over the package's own docstrings, so a number written in prose is executable. Halving an exponent used to leave the suite green	499 doctests inside 3904 tests

THE ONE PLACE THE HONEST ANSWER IS NARROWER THAN THE BRIEF

Objective 2 asks for unauthorized verification attempts to be detected. Verification consumes exactly one resource, a recipient's measurement record bound to a distribution round, so the threat is really three threats indexed by how an outsider gets one, and they have different answers.

ROUTE	HOW HE GETS A RECORD	THE ANSWER
Off the wire	measures the Phase A qubits in flight	Detected, 40/40 . No-cloning forbids copying, so he has to measure, and measurement disturbs. This is intercept-resend under another name and the channel screen already catches it
Invented	fabricates the entries himself	Refused at the interface, and answered by proof rather than by a rate. His mismatch sits at 0.4994 against Alice's genuine declaration and 0.5052 against one she never made, intervals overlapping, so his verdict is independent of whether the signature is real. There is no successful verification left to detect
Stolen	holds a real recipient's log, taken outside the protocol	Excluded by assumption (RECORD SECRECY), beside (AUTH) in the same table. He runs the identical computation the owner runs on the owner's data, and no transcript field differs. Separating him would mean separating two evaluations of one pure function on one argument, so no protocol catches this, not just this one

Two things that reading leaves out, and both are stated wherever the claim appears in the repository. No detector fires on an unauthorized verification attempt as such. And the authorised argument that refuses the fabricator is off by default, is never passed by the shipped session, and produced no published number here, so it is an interface a deployment could use rather than a defence this one runs.